

INDEX

Content		Page Number
I.	Executive Summar	01
II.	Project Goals & Objectives	02
III.	Proposed Solution and Methodology	04
IV.	Scope of Work (SoW)	10
V.	Project Timeline & Milestones	14
VI.	Technical Architecture	17
VII.	Prerequisites, hardware & resource planning	23
VIII.	Project Team & Management	26
IX.	Quality Assurance (QA) & Testing	30
X.	Risk Management	31
XI.	Commercial & Investment	32
XII.	Maintenance & Support	32
XIII.	Company Profile	34

I. EXECUTIVE SUMMARY

The proposed solution provides a comprehensive framework for the Mission and Combat System Research and Design Centre (MCSRDC) at HAL to modernize its procurement through an automated Smart Clause Selection Tool. The system employs a hybrid intelligence approach, combining AI and Natural Language Processing (NLP) for intelligent document interpretation with deterministic rule-based logic to ensure absolute policy compliance and consistency.

Strategic Value and Modernization

- **End-to-End Automation:** The tool creates a unified software solution for the entire RFQ lifecycle, automating the generation of Annexure-1 (Technical), Annexure-2A/2B (Commercial), Annexure-3 (MII Compliance), and Annexure-4 (Price Bid).
- **Operational Excellence:** By streamlining the preparation of tender and contract documentation, the solution significantly reduces manual effort, eliminates human error, and ensures strict adherence to standardized organizational templates.
- **Policy Governance:** The architecture is designed so that mandatory policy clauses and financial thresholds always take precedence over AI-assisted suggestions, maintaining the high level of governance required by HAL.

Security and Data Sovereignty

- **Secure Onsite Deployment:** Recognizing the sensitive nature of defence procurement data, the application is a web-based solution hosted entirely within the HAL internal network.
- **Air-Gapped Operation:** All data processing, storage, and AI operations remain within the HAL network environment with zero dependency on external cloud services or internet connectivity.
- **Data Control:** By hosting the system internally, HAL maintains absolute ownership and physical control over its information assets, ensuring no data ever leaves the organization's physical environment.

- Reliability and Long-Term Commitment
- Human-in-the-Loop: No annexure or clause is finalized without explicit review and confirmation by an authorized HAL user, ensuring total auditability and trust.
- Auditability and Traceability: The system maintains a permanent audit trail of all user actions, AI suggestions, and rule validations, providing total transparency for internal reviews and statutory compliance.
- Sustainable Partnership: The implementation is led by a specialized onsite technical team, followed by a minimum of 10 years of stabilization and technical support to ensure the tool's longevity and reliability.

I. PROJECT GOALS AND OBJECTIVES

- The primary objective of the Smart Clause Selection Tool is to transform the complex, manual process of RFQ (Request for Quotation) preparation into a streamlined, high-accuracy digital workflow. The project is designed to bridge the gap between unstructured technical requirements and the structured, governed world of HAL procurement policy.

2.1 Core Project Goals

- **Total Policy Compliance:** To eliminate the risk of missing mandatory clauses by enforcing a "Rule-First" architecture where the HAL Purchase Manual and internal circulars are programmatically applied to every tender.
- **Process Standardization:** To ensure that all annexures (1, 2A/2B, 3, and 4) generated by MCSRDC follow a single, standardized format, regardless of the individual officer handling the file.
- **Zero Data Leakage:** To operate entirely within an air-gapped, onsite environment, ensuring that sensitive defence procurement data remains within HAL's physical and digital perimeter.
- **Significant Lead-Time Reduction:** To reduce the time taken to prepare a consolidated RFQ package from several hours/days to **under 30 minutes**, allowing officers to focus on strategic vendor evaluation.

2.2 Key Technical Objectives

The project will deliver against these specific technical benchmarks:

A. Intelligent Parsing & Extraction

- Automate the extraction of metadata and technical specifications from unstructured PDF and Word documents for **Annexure-1**.
- Utilize local Natural Language Processing (NLP) to identify key technical parameters and map them to the corresponding commercial terms.

B. Contextual Clause Retrieval (RAG)

- Implement a **Retrieval-Augmented Generation (RAG)** framework that allows users to search the Master Clause Library using natural language queries (e.g., *"Show clauses for indigenous vendor development"*).
- Ensure that retrieved clauses are contextually relevant to the specific tender's value, location, and bidder type.

C. Deterministic Rule Enforcement

- Build a "Hard-Coded" Rule Engine that validates AI suggestions against mandatory thresholds (e.g., automatically applying specific "Make in India" clauses based on order value).
- Provide a "Human-in-the-Loop" validation screen where users can see exactly *why* a specific clause was suggested or enforced.

D. Seamless Document Synthesis

- Develop an output engine capable of generating consolidated ZIP packages containing Excel files that preserve HAL's native formulas and formatting.
- Integrate with existing data sources (IFS ERP/CSV exports) to fetch financial and item level data for **Annexure-4 (Price Bids)**.

2.3 Success Metrics (KPIs)

The success of the deployment will be measured against the following Key Performance Indicators:

1. **Accuracy:** The system shall achieve a minimum of 90% accuracy in clause identification as per RFQ requirements. Additionally, each AI-generated clause shall include a confidence score indicating the reliability of the prediction. Clauses with lower confidence scores shall be highlighted for user review and validation.
2. **Efficiency:** >75% reduction in the time required to compile a complete RFQ annexure set.
3. **Compliance:** 100% adherence to the mandatory clauses defined in the Master Clause Library.
4. **Adoption:** 100% of new tenders at MCSRDC processed through the tool post-stabilization phase.

III. PROPOSED SOLUTION AND METHODOLOGY

The proposed solution is a secure, web-based application designed to automate end-to-end tender document preparation and contract clause selection for HAL. It utilizes a **hybrid approach** that integrates AI and Natural Language Processing (NLP) with deterministic, rule based logic to ensure accuracy while maintaining strict governance.

3.1 Overview of the Proposed System

The system acts as a centralized platform for creating standardized Request for Quotation (RFQ) packages. By combining document intelligence with HAL-specific procurement rules, the solution ensures that all interpretations of technical requirements remain compliant with organizational policies.

3.2 Key Functional Capabilities

The system automates the RFQ lifecycle through several core functions:

- **Automated Annexure Generation:** Produces Annexures 1, 2A/2B, 3, and 4 in HAL approved Excel formats with parallel PDF outputs.

- **Clause Selection & Applicability:** Automatically identifies and selects relevant commercial clauses from a governed Master Clause Library based on validated tender parameters.



- **Document Interpretation Support:** Extracts metadata and requirements from unstructured PDF and Word documents to prepopulate annexures.
- **Consolidated RFQ Package Creation:** Bundles all finalized annexures and mandatory enclosures into a single package for distribution.
- **Enclosures Processing:** Automatically identifies and attaches supporting documents, such as policy circulars and specifications, based on the selected clauses and tender context.

3.3 Technical Architecture

The tool is built on a modular, layered architecture to ensure secure operation within the HAL intranet:

- **User Layer:** Facilitates document uploads, natural-language query entry, ERP/CSV data connection, and final package downloads.
- **Application Layer:** Manages user sessions, authentication, and guided workflows for clause review.
- **Processing Layer:** Houses the **Orchestrator, AI/NLP Engine** for document parsing, and the **Rule Engine** for policy enforcement.
- **Data Layer:** Contains the Master Clause Library, Relational Database, Vector Store for AI, and HAL templates.
- **Output Layer:** Delivers finalized, HAL-compliant documents.

3.4 Structured Workflow & Governance

The system follows a five-step workflow designed to ensure accuracy and human accountability:

1. **Input Collection:** Users upload technical specs and provide tender parameters via natural-language queries.
2. **Document Interpretation:** The AI engine parses technical requirements while the Rule Engine evaluates policy conditions.
3. **Clause Mapping and Review:** Clauses are mapped from the Master Library; users review selections, including mandatory policy inclusions.
4. **Template Population:** Data is automatically populated into approved templates, preserving existing formulas and structural integrity.
5. **Final Packaging:** Documents are consolidated into a controlled RFQ package for export.

3.5 Control Mechanisms

To maintain defense-grade auditability, the methodology incorporates four primary controls:

- **Validation Checkpoints:** Automated checks prevent proceeding if mandatory information is missing or if conflicts are detected.

- **Rule Enforcement:** Deterministic rules ensure that mandatory statutory language cannot be altered by the AI or user.
-

- **Human-in-the-Loop:** AI is a decision-support tool only; no annexure is finalized without explicit authorization from a HAL user.
- **Audit and Traceability:** All actions, decisions, and approvals are permanently recorded in audit logs.

3.6 Data Flow and Control Mechanism

To ensure defence-grade accountability, consistency, and auditability, the methodology incorporates a rigorous data flow managed by four primary control layers:

- **Validation Checkpoints:** Automated checks verify the completeness of all inputs at each stage. The system prevents progression if it detects conflicts or missing mandatory information, ensuring no incomplete tender is processed.
- **Deterministic Rule Enforcement:** A core Rule Engine ensures that mandatory policy clauses and statutory language—derived from the HAL Purchase Manual and applicable circulars—cannot be altered or excluded by AI suggestions.
- **Human-in-the-Loop (HITL) Control:** AI serves strictly as an assistive layer. No document, technical requirement, or commercial clause is finalized without explicit review and manual confirmation by an authorized HAL user.
- **Conflict Resolution:** Any AI-derived suggestion that conflicts with established policy thresholds or mandatory rules is automatically rejected by the system before it reaches the user. Cases that are ambiguous or conditional are flagged specifically for human intervention.
- **Audit and Traceability:** Every user action, AI suggestion, rule validation, and manual approval is recorded in permanent audit logs. These logs include version history and timestamps to provide total transparency for future internal reviews or statutory audits.
- **Confidence-Based Validation:** Each AI-generated clause is assigned a confidence score. Clauses with low confidence are automatically flagged for user review. The

system also maintains traceability by linking each clause to its source within the Master Clause Library or policy documents.

- **Continuous Learning Mechanism:** The system shall capture user feedback such as clause approvals, modifications, and rejections, and utilize this information to continuously improve the clause selection model, enabling adaptive and improved accuracy over time.

3.7 Master Clause Library Governance

The integrity of the "Proposed Solution" relies on the **Master Clause Library**, which acts as the single source of truth for all generated outputs.

- **Governed Sourcing:** Clauses are sourced exclusively from authorized HAL documents, such as the Purchase Manual and Delegation of Powers.
- **Data Pre-processing:** Source documents undergo a structured cleaning process to remove non-relevant content (headers/footers) and segment policy text into individual, standardized clauses.
- **Metadata Tagging:** Each clause is stored with metadata defining its status as Mandatory, Optional, or Conditional, as well as its applicability based on bidder type or contract value thresholds.
- **Administrative Oversight:** While AI may assist in identifying clauses from new policy documents, only authorized HAL administrators can modify or activate clause content within the library

3.8 Scalability & Future-Proofing

While the initial deployment is optimized for the specific requirements of MCSRDC, the Smart Clause Selection Tool is designed as an **enterprise-ready foundation**. The architecture ensures that as HAL's procurement needs and the broader technological landscape evolve, the system can scale without requiring a complete redesign.

3.8.1 Enterprise-Wide Expansion

- **Division Support:** The system's modular design allows for the creation of "Division-Specific Profiles." While the core engine remains the same, other

HAL divisions (e.g., LCA, Aircraft, or Engine Division) can plug in their own localized Master Clause Libraries and technical templates.

- **Centralized Governance, Decentralized Use:** The tool can be scaled to a centralized server model where multiple departments access the same secure internal web portal, ensuring standardized procurement language across the entire organization.

3.8.2 Adaptability to Policy Changes

- **Dynamic Rule Updates:** The Rule Engine is decoupled from the core code. When the HAL Purchase Manual or "Make in India" (MII) thresholds change, administrators can update the logic parameters via a secure UI without needing to rewrite software.
- **Version-Controlled Clause Library:** As new statutory mandates are issued, they can be uploaded to the Master Library as "Active" versions, while archived clauses are retained for reference in older, ongoing contracts.

3.8.3 Technological Evolution & AI Portability

- **Model Agnostic Architecture:** The system is built to be "Model Agnostic." As more powerful, efficient LLMs (Large Language Models) are released (e.g., future versions of Llama or specialized Indian Defence LLMs), the Processing Layer can be upgraded with minimal disruption.
- **Vector Database Scaling:** The use of a Vector Database (Chroma DB/Qdrant) ensures that the system can handle a library of tens of thousands of clauses with sub-second retrieval speeds, ensuring the tool remains fast even as its data grows.

3.8.4 Increased Document Complexity & Volume

- **Handling High-Volume Tenders:** The architecture supports distributed processing. If the volume of RFQs increases significantly, additional GPU nodes can be added to the internal cluster to handle parallel document parsing for multiple users simultaneously.
- **Extensible Annexure Modules:** The framework is designed to accommodate future requirements beyond Annexures 1-4. New modules for Service Level

Agreements (SLAs), Non-Disclosure Agreements (NDAs), or Post-Contract Management can be integrated into the existing workflow.

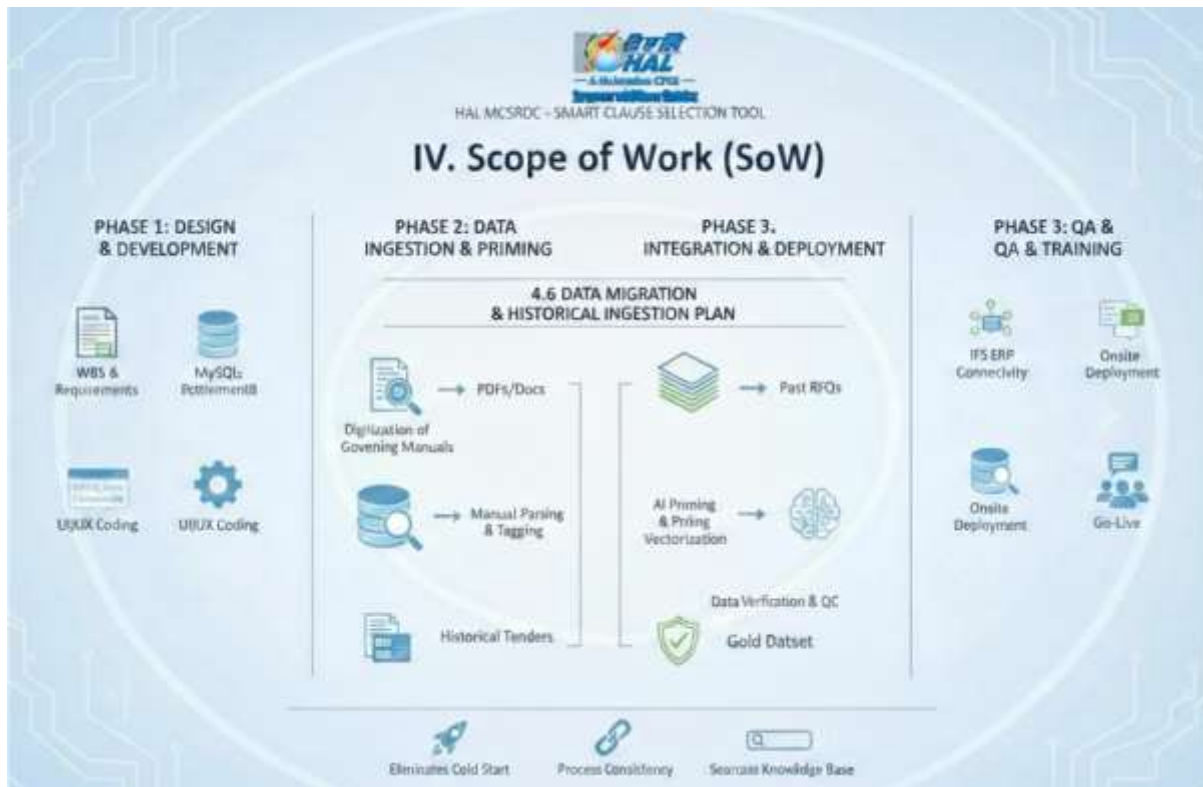
II. SCOPE OF WORK (SOW)

The scope of work involves the end-to-end development, onsite deployment, and long-term maintenance of the Smart Clause Selection Tool, specifically tailored to handle the high complexity documents utilized by HAL (MCSRDC).

4.1 Annexure-1: Technical Requirement Parsing

The system must transform unstructured technical specifications into structured, HAL compliant Excel documents.

- **Multi-Format Ingestion:** Support for PDF and Word documents as primary inputs.
- **Intelligent Parsing:** AI-driven extraction of text, complex technical tables, and embedded images.
- **Structural Mapping:** Accurate mapping of extracted content into corresponding sections of the HAL technical Excel template.
- **Integrity Control:** Generation of an Excel (XLS/XLSM) output where buyer requirements are locked, and only bidder response fields remain editable.



4.2 Annexure-2A/2B & Annexure-3: Commercial and Policy Compliance

The system automates the selection of commercial terms and "Make in India" (MII) formats based on validated tender parameters.

- **Parameter Derivation:** Processing of natural-language queries to identify tender type, procurement mode, and bidder category.
- **Clause Selection:** Automated retrieval of applicable clauses from the Master Clause Library based on predefined rule sets.
- **MII Logic:** Deterministic selection of MII templates based on policy thresholds and tender category, ensuring no modification of approved policy text occurs.
- **Dual-Output Generation:** Production of both HAL-approved Excel formats and issuance ready PDF documents.

4.3 Annexure-4: Price Bid Integration

Automating the financial offer structure while maintaining mathematical integrity.

- **ERP/Data Integration:** Extraction of line-item data via secure CSV uploads or direct IFS ERP interfaces.

- **Formula Preservation:** Mapping line items into templates while preserving existing HAL formulas, calculations, and formatting.
- **Evaluation Integrity:** Protection of cells to ensure bidder submission uniformity and evaluation accuracy.

4.4 RFQ Package Consolidation and Enclosures

The final stage of the scope involves compiling all disparate documents into a unified package.

- **Enclosure Identification:** Identification of mandatory supporting documents (policy circulars, reference documents) based on selected clauses.
- **Unified Packaging:** Consolidation of all annexures and enclosures into a single, version controlled ZIP package for distribution.
- **Naming Conventions:** Application of HAL-defined versioning and naming standards for auditability.
- **Enclosure Management:** The system shall provide an administrative interface to manage the enclosure repository, enabling authorized users to add, update, delete, and replace documents such as Bank Guarantee formats, statutory declarations, and integrity pact documents.

4.5 Implementation and Support Responsibilities

- **Onsite Development:** Full execution of design, development, and configuration at the MCSRDC facility in Bangalore.
- **Training:** Delivery of end-user and administrator training, including the provision of technical manuals.
- **Extended Maintenance:** Provision of stabilization support and long-term technical maintenance for a minimum of 10 years.

4.6 Data Migration & Historical Ingestion Plan

To ensure the Smart Clause Selection Tool is effective from "Day 1," a structured data ingestion process is required. This phase focuses on populating the **Master Clause Library** and "priming" the AI engine with HAL's historical procurement data to ensure the language model understands the specific technical context of MCSRDC.

4.6.1 Digitization of Governing Manuals

The system's "Ground Truth" is derived from HAL's official policy documents. The ingestion process includes:

- **Manual Parsing:** Systematic extraction of clauses from the *HAL Purchase Manual*, *Delegation of Powers (DoP)*, and relevant *Internal Circulars*.
- **Clause Segmentation:** Breaking down dense policy chapters into individual, searchable "Clause Units" with unique IDs.
- **Metadata Tagging:** Manually assigning attributes to each clause (e.g., "Mandatory for Global Tenders," "Applicable for orders > 50 Lakhs," or "Specific to MII Compliance").

4.6.2 Historical Tender Ingestion (AI Priming)

To improve the accuracy of **Annexure-1 (Technical Specs)** and **Annexure-4 (Price Bids)**, the system will ingest historical data:

- **"Perfect Tender" Baseline:** Ingesting the last 2–3 years of successfully issued RFQs to serve as high-quality examples for the AI.
- **Technical Vocabulary Training:** Extracting technical parameters from previous MCSRDC projects so the AI recognizes specialized aviation and electronic warfare terminology.
- **Template Mapping:** Analyzing how past unstructured technical requirements were manually mapped into Excel templates to automate that logic in the new system.

4.6.3 Data Verification & Cleaning

Before data is committed to the production **Vector Store**, it undergoes a quality control (QC) gate:

- **De-duplication:** Ensuring that redundant or outdated versions of clauses are removed or marked as "Archived."
- **Formatting Standardization:** Converting various source formats (scanned PDFs, old Word docs, etc.) into a clean, structured JSON format for the database.
- **HAL Review:** A final "Gold Dataset" will be presented to HAL administrators for verification before the system goes live.

4.6.4 Support from HAL

The successful implementation of the proposed solution will be supported by HAL through:

- Provision of development workstations within the HAL network environment (air-gapped setup)
- Access to procurement policies, templates, and relevant documentation
- Availability of historical tender and contract data for system configuration and AI training
- Functional guidance and validation support from domain experts
- Nomination of responsible personnel for requirement clarification, testing, and UAT sign-off
- Controlled internet access via separate systems, as per HAL IT policies

V. PROJECT TIMELINE AND MILESTONES

The project is structured into eight distinct phases over a total estimated duration of **20–28 weeks**, followed by long-term stabilization and maintenance. Each phase concludes with a formal sign-off to ensure alignment with HAL's technical and security requirements.



PROJECT TIMELINE & MILESTONES

Phase 1

Project Initiation & Requirements



Kick-off meeting; finalization of functional and non-functional requirements; confirmation of HAL-approved templates

Weeks 1-3



Requirement Sign-off

Phase 2

System Design & Architecture



Finalization of workflows, data models, architecture, and Master Clause Library structure

Weeks 4-6



Design Approval

Phase 3

Development & Configuration



Application development; AI/NLP configuration; rule-based logic implementation; Master Clause Library setup with continuous QA validation

Weeks 7-16



Development Completion

Phase 4

System Integration & Testing



Module integration; validation of Excel, PDF, and Word outputs; access control and security verification

Weeks 17-20



System Testing Sign-off

Phase 6

User Acceptance Testing (UAT)



Execution of UAT scenarios by HAL users; ont of incorporation of approved refinements

Weeks 22-20



Design Approval

Phase 6

Deployment & Installation



Onsite deployment within HAL network; configuration of production users and roles

Weeks 26-25



Go-Live Approval

Phase 6

Training & Knowledge Transfer



End-user and administrator training; handover of user manuals and technical documentation

Ongoing



Project Closure

5.1 Project Phase Breakdown



5.2 Key Milestone Deliverables

M1: Requirements & Infrastructure Sign-off

Finalization of the Master Clause Library structure and the specific data points required from technical documents for Annexure-1. Hardware procurement and preparation of the onsite server environment at MCSRDC.

M2: Core Logic & AI Engine Validation

Successful demonstration of the AI/NLP engine's ability to parse complex technical tables from sample HAL documents and the Rule Engine's ability to enforce mandatory "Make in India" thresholds.

M3: End-to-End Annexure Integration

Validation that all four annexures (1, 2A/2B, 3, and 4) can be generated as a single, consolidated ZIP package with consistent metadata and formatting.

M4: User Acceptance (UAT) & Production Handover

Exhaustive testing by HAL procurement officers using real-world past tenders to verify selection accuracy and system stability. This culminates in the final handover of the production environment.

Post Go-Live Support

The system shall include a post go-live support phase of 3 months following deployment, training, and user acceptance to ensure system stabilization and issue resolution.

5.3 Resource Deployment Schedule

To meet these milestones, the following onsite team will be deployed:

Weeks 1–6: Focus on Requirements, Design, and Infrastructure setup.

Weeks 7–18: Intensive development, AI model fine-tuning, and rule-based logic configuration.

Weeks 19–Final: Testing, Deployment, and User Training sessions.

VI. Technical Architecture

The Technical Architecture of the Smart Clause Selection Tool is designed for high-performance, onsite execution within the HAL MCSRDC intranet. It follows a decoupled



service-oriented structure to ensure that AI processing, rule enforcement, and document generation remain efficient and secure.

6.1 Multi-Layered System Architecture

The application is organized into five logical layers, ensuring that data flows from raw input to finalized RFQ packages through controlled gates.

- **User Layer (Interface):** A responsive web portal accessible via HAL's internal network. It provides specialized dashboards for Procurement Officers to upload documents, input natural-language queries, and review generated clauses.
- **Application Layer (Governance):** Manages the "Human-in-the-Loop" workflows. This layer handles user authentication, session management, and the state-machine that tracks a document from "Draft" to "Approved."
- **Processing Layer (Intelligence & Logic):**
 - * **AI/NLP Engine:** Utilizes Large Language Models (LLMs) like **Llama 3.2** or **Mistral** (deployed locally) for technical text extraction and semantic understanding of natural language queries.
 - * **Rule Engine:** A deterministic logic gate that maps tender parameters (e.g., contract value, bidder type) to mandatory clauses. It acts as the final authority over AI suggestions.
 - * **Orchestrator:** Coordinates data movement between the AI engine and the Rule Engine.
- **Data Layer (Storage):**
 - * **Relational Database (MySQL/PostgreSQL):** Stores structured metadata, audit logs, and user permissions.
 - * **Vector Database (ChromaDB/Qdrant):** Stores the Master Clause Library as mathematical embeddings to enable rapid, context-aware clause retrieval (RAG).
 - * **Template Store:** A secure repository of HAL-approved Excel and Word templates.
- **Output Layer (Generation):** A dedicated document assembly engine that populates Excel files, preserves formulas, and converts final versions to PDF.
- The proposed architecture also aligns with modular components to ensure scalability and maintainability:
 - Document Ingestion Module for handling technical documents, policy manuals, and templates in PDF/Word formats
 - Clause Extraction and Processing Engine utilizing NLP/AI techniques for structured clause extraction
 - Master Clause Library for centralized storage and governance of clauses
 - Smart Clause Selection Engine combining rule-based logic and AI-assisted recommendations

- Template Generation Module for creating RFQ annexures as per HAL templates
- Integration Layer for ERP/CSV-based data exchange
- User Interface & Administration Module for user interaction, approvals, and system configuration

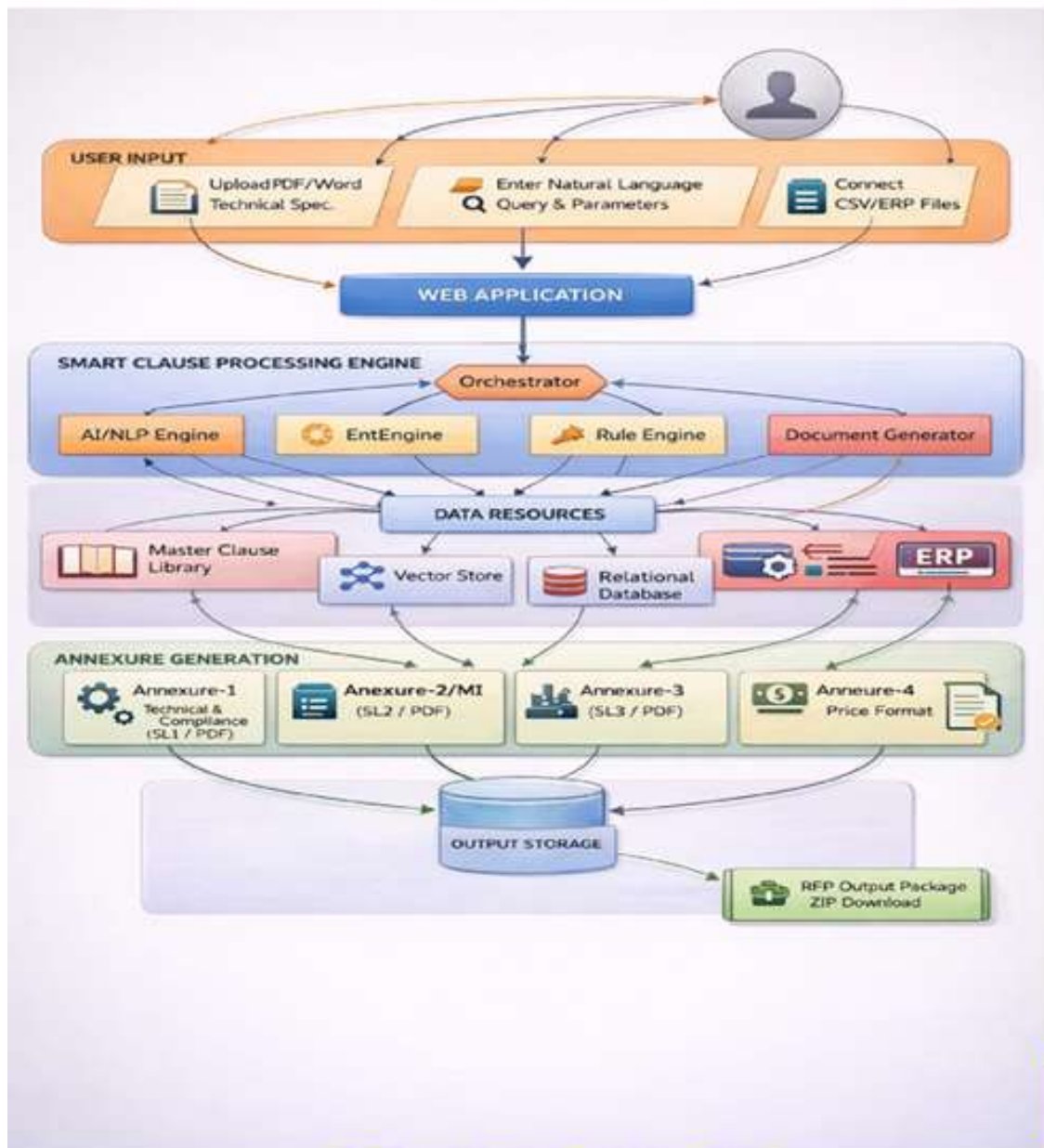
6.2 The Hybrid AI & RAG Framework

To achieve high accuracy without internet dependency, the system employs **Retrieval Augmented Generation (RAG)**.

-
1. **Ingestion:** The Master Clause Library is "chunked" and stored in the local Vector Database.
 2. **Retrieval:** When a user enters a query or uploads a spec, the system retrieves the most relevant policy clauses from the internal library.
 3. **Augmentation:** These clauses are provided to the local LLM as "ground truth" to prevent AI hallucinations.
 4. **Generation:** The LLM drafts the annexure content based *only* on the provided library text.

6.3 Security & Integration Architecture

- **Air-Gapped Operation:** The entire stack—from the OS to the AI models—is installed via offline media or controlled local mirrors.
 - **ERP Connectivity:** The system interfaces with HAL's **IFS ERP** via secure CSV export/import or a dedicated internal API to fetch financial line items for Annexure-4.
 - **Encryption:** All data at rest in the MySQL and Vector databases is encrypted using industry-standard protocols.
-



safeguard against hardware failures, data corruption, and operational downtime within the airgapped HAL environment.

- **High-Availability (HA) Configuration:** The system supports a "Hot Standby" or redundant server configuration. If the primary application server fails, traffic is rerouted to the secondary node to ensure uninterrupted procurement workflows.
- **Fail-Safe "Rule-Only" Mode:** In the event of a GPU hardware failure, the system is designed to switch to a degraded "Rule-Only" mode. While AI-

assisted document parsing may be disabled, the deterministic Rule Engine remains functional, allowing users to continue generating compliant annexures using manual parameter inputs.

- **Automated Local Backups:** All critical data—including the Master Clause Library, the Vector Store (ChromaDB), and the Relational Database (MySQL)—is subject to automated daily backup procedures. These backups are stored on separate HAL-approved storage nodes to prevent data loss from a single point of failure.
- **Data Integrity & Recovery:** The system maintains full version control and audit logs for every document and clause. In the event of accidental data corruption, administrators can restore the Master Clause Library to a previous known-good state using the timestamped version history.
- **Onsite Support & Stabilization:** A dedicated Deployment/MLOps Engineer will be available to manage onsite recovery procedures and hardware optimization, ensuring that the system's performance meets the 15–30 second RFQ generation target even after a recovery event.

6.3 Indicative Technology Stack

The proposed solution is built using industry-standard, scalable, and secure technologies aligned with HAL requirements:

- **Application Layer:** Python-based frameworks (FastAPI/Django) for backend services and web application development
- **AI/NLP Processing:** HuggingFace Transformers, spaCy, and PyTorch/TensorFlow for document understanding and clause extraction
- **Database Layer:** PostgreSQL/MySQL for structured data and ChromaDB/Qdrant for vector-based clause retrieval
- **Document Processing:** Apache Tika, Tesseract OCR, and Python-based libraries for PDF and Word document parsing
- **Integration Layer:** REST APIs and secure CSV-based interfaces for ERP (IFS) integration
- **Deployment Environment:** On-premise Linux/Windows server within HAL network, ensuring no dependency on external cloud services

VII. PREREQUISITES, HARDWARE & RESOURCE PLANNING

This section outlines the technical environment, high-performance infrastructure, and specialized personnel required for the onsite development and long-term operation of the **Smart Clause Selection Tool** within the HAL (MCSRDC) network.

7.1 System Prerequisites & Hardware

The solution is engineered for a hybrid lifecycle: an internet-enabled development phase to leverage modern AI frameworks, followed by a strictly on-premise, air-gapped deployment for production security.

- **Development Workstations (Connectivity Enabled):**
 - **Internet Facility:** Required during the development phase to securely fetch open-source libraries, AI model weights (LLaMA 3 / Mistral), and framework updates.
 - **AI-Augmented Development:** The team will utilize the Cursor AI code editor to accelerate code generation, automate debugging, and optimize the integration of RAG and GenAI components.
 - **Hardware:** Intel i7 (12th Gen+) / Xeon-class processor with 32–64 GB RAM and 1 TB NVMe SSD to support local containerized development and testing of vLLM with small-to-medium models (7B–8B).
- **Application Deployment Server (HAL Data Centre):**
 - **GPU Requirement (Critical):** To ensure the system meets the performance target of generating full RFQ packages within 15–30 seconds, a high-memory GPU such as NVIDIA A100 (40GB/80GB), RTX 6000 Ada (48GB), or equivalent is required for efficient vLLM-based inference. Multi-GPU configuration (2–4 GPUs) is recommended for higher throughput and scalability.

- **Processing Power:** Dual Intel Xeon or AMD EPYC (16–32 cores recommended) to manage API services, rule engine execution, and orchestration of inference workloads.
 - **Memory:** Minimum 128 GB RAM (expandable to 256 GB) to support concurrent users, KV cache handling, and RAG pipeline operations.
 - **Storage:** Minimum 2 TB NVMe SSD for fast model loading, embedding storage, logging, and retrieval operations.
 - **Network:** High-speed internal network (10 Gbps recommended) to ensure low-latency communication between application components.
 - **Platform & Deployment:** Linux-based environment (Ubuntu 20.04 or higher), Docker for containerization, and Kubernetes (optional) for scaling and high availability.
 - **Scalability & High Availability:** Support for horizontal scaling using multiple GPU nodes, load balancing for concurrent requests, and failover mechanisms to ensure uninterrupted service.
- **Storage & ERP Integration:**
 - **Database:** Dedicated servers for **MySQL/PostgreSQL** (Master Clause Library) and **ChromaDB/Qdrant** (Vector Store).
 - **Integration:** Secure CSV-based data exchange or approved API interfaces with the **IFS ERP system** for Annexure-4 financial data extraction.

7.2 Specialized Onsite Team Structure

To ensure successful delivery and decade-long stability, C2C Advanced Systems will deploy a 5-person specialist team onsite at the MCSRDC facility.

Role	Responsibility
Solution Architect	System design, security alignment, and integration with HAL's internal IT ecosystem.
AI/ML Engineers (2)	Local LLM configuration (Llama 3.2/Mistral), RAG logic development, and technical table parsing.
QA/Testing Engineer	Functional and security testing; ensuring the Rule Engine correctly overrides AI suggestions.
Deployment/MLOps Engineer	Onsite server installation, air-gapped maintenance, and long-term stabilization support.

7.3 Security & Environment Isolation

- **Air-Gapped Operation:** The entire software stack, including the OS and AI models, will be installed via offline media or controlled local mirrors to maintain zero dependency on external networks.
- **Data Sovereignty:** All data processing and AI operations remain within the HAL perimeter, ensuring absolute physical control over sensitive defense procurement information.
- **Encrypted Storage:** All data at rest in the relational and vector databases will be encrypted using industry-standard protocols.

7.3 Business Continuity & Redundancy

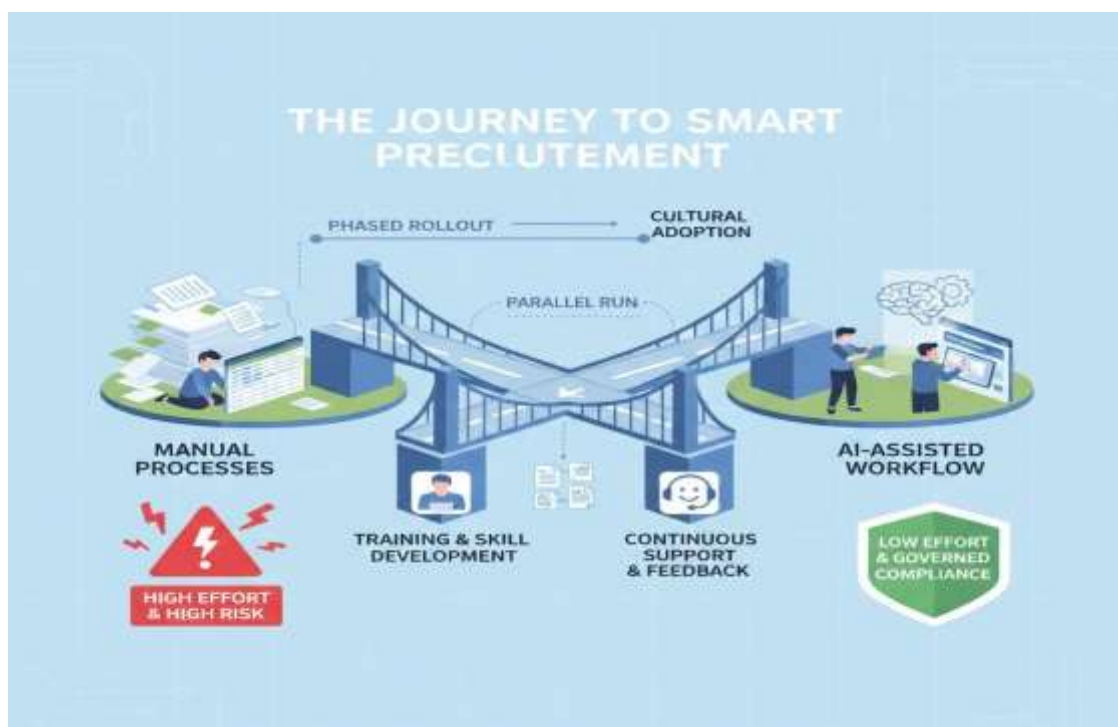
- **High-Availability (HA):** The architecture supports a "Hot Standby" configuration to ensure uninterrupted procurement workflows in case of primary server failure.
- **Fail-Safe Mode:** If GPU hardware fails, the system switches to a "Rule-Only" mode, allowing compliant annexure generation via manual parameter inputs.
- **Automated Backups:** Daily automated backups of the Master Clause Library and audit logs will be stored on separate HAL-approved storage nodes.

7.4 Security, Licensing, and Documentation Compliance

- The solution shall be deployed as a secure, on-premise application operating entirely within the HAL network environment
- All software components, libraries, and frameworks used in the solution shall be properly licensed for commercial use
- In case of open-source components, licensing terms shall be adhered to and documented appropriately
- The system shall be delivered with complete documentation including User Manuals, Administrator Guides, and Architecture Documents
- The solution shall comply with security standards and ensure that no malicious code is present in the deployed system

VIII. PROJECT TEAM AND MANAGEMENT

To ensure the successful delivery and long-term stability of the Smart Clause Selection Tool, a dedicated onsite team of specialists will be deployed at the MCSRDC facility. This team is structured to manage the complete lifecycle from initial AI training to decade-long maintenance.



- **Solution Architect (1 Role):**

- **Focus:** Overall system design, security alignment, and integration with HAL's internal IT ecosystem.
- **Experience:** 8+ years in enterprise software architecture and secure network deployments.

8.1 Responsibility Matrix

Responsibility Matrix		
HAL (MCSRDC)	Key Responsibilities	
	• Provide secure workstations and controlled network access within HAL premises.	
	• Provide server and infrastructure resources for hosting the application.	
	• Supply HAL-approved RFQ templates, procurement manuals, and the policy circulars	
	• Provide domain expertise and historical data for configuration and validation.	
	• Participate in requirement clarification design reviews, and User Acceptance Testing (UAT).	
	• Nominate Subject Matter Experts (SMEs) for project coordination	
The Bidder		
	• Deploy qualified technical resources (Architect, AI Engineers, QA, MLOps) onsite at Bangalore	
	• Design, develop, and implement rule-based logic strictly as per HAL procurement policies	
	• Configure AI/NLP on the Master-based logic strict library using approved source documents policies	
	• Conduct end-user training and provide technical documentation/manuals	
	• Provide post-go-live support and long-term stabilization	
Operational Coordination		
	• Progress Reviews: Periodic meetings to track progress...	
	• Requirement Validation: Joint reviews confirm correct implementation...	
	• Testing and UAT: Coordinated functional testing...	

8.2 Specialized Onsite Team Structure

The team consists of five key roles, each providing a specific layer of expertise required for a secure, defence-grade deployment:

- **AI/ML Engineers (2 Roles):**
 - **Focus:** Configuring the local LLM (Llama 3.2/Mistral), developing the Retrieval Augmented Generation (RAG) logic, and training the engine to parse complex technical tables from Annexure-1.
 - **Experience:** 3+ years in NLP, vector databases, and document intelligence.
- **QA/Testing Engineer (1 Role):**
 - **Focus:** Executing rigorous functional, security, and performance testing. Ensures that the deterministic Rule Engine correctly overrides AI suggestions in every scenario.
 - **Experience:** 4+ years in software quality assurance and security verification.
- **Deployment & MLOps Engineer (1 Role):**
 - **Focus:** Managing onsite server installation, air-gapped environment maintenance, and providing the **10-year stabilization support**.
 - **Experience:** 5+ years in DevOps, server management, and system administration.

8.3 Project Management Governance

The project will follow a structured governance model to ensure transparency and adherence to timelines:

- **Weekly Progress Reviews:** Joint meetings with HAL stakeholders to review milestone achievements and address technical hurdles.
- **Documented Sign-offs:** Each phase (as detailed in the Timeline) requires a formal signoff from HAL before proceeding to the next.

- **Risk Management Office:** A dedicated track to identify potential policy changes in the HAL Purchase Manual and update the Master Clause Library accordingly

8.4 Change Management & Adoption Strategy

The transition from a manual, document-heavy procurement process to an AI-assisted workflow represents a significant operational shift for the MCSRDC team. To ensure high user adoption and minimize resistance, the following framework will be implemented:

8.4.1 The "Parallel Run" Adoption Phase

To build trust in the AI's document interpretation and the Rule Engine's clause selection, the system will be deployed using a "Parallel Run" strategy during the first 4–6 weeks:

-
- **Dual Preparation:** Users will prepare a set of "Shadow Tenders" using the tool while continuing their standard manual process.
 - **Accuracy Benchmarking:** Outputs from the Smart Clause Selection Tool will be compared against the manual versions. This allows users to see firsthand the tool's ability to catch missed clauses or formatting errors.
 - **Gradual Transition:** Once a 98% accuracy threshold is consistently met, the manual requirement will be phased out.

8.4.2 Super-User "Champion" Program

We will identify and train "**Super-Users**" within the procurement and technical departments.

-
- **Role:** These individuals will receive advanced training on the Master Clause Library and the Rule Engine logic.
 - **Benefit:** They act as first-level support for their peers, ensuring that minor functional questions are resolved immediately onsite, which reduces frustration and speeds up adoption.

8.4.3 Feedback Loop & Continuous Improvement

A technology-driven tool must feel responsive to user expertise. The system includes a "**Flag & Feedback**" module:

-
- **User Overrides:** If a user modifies an AI-suggested clause, they can "Flag" the reason (e.g., "Outdated policy" or "Contextual exception").
 - **Monthly Review:** During the stabilization phase, our onsite MLOps Engineer will review these flags with HAL leadership to update the Rule Engine or the Vector Store, ensuring the tool "learns" from HAL's expert procurement officers.

8.4.4 Phased Training Curriculum

Training will be tailored to specific user roles to prevent information overload:

- **For Technical Officers:** Focus on Annexure-1 (technical spec parsing) and metadata extraction.
- **For Procurement/Commercial Officers:** Focus on Annexure-2/3, clause applicability logic, and MII compliance.
- **For Administrators/IT:** Focus on the "Data Layer," Master Library updates, and system health monitoring.

IX. QUALITY ASSURANCE (QA) AND TESTING

The QA framework for this tool is designed to eliminate the risks typically associated with AI (such as "hallucinations") and ensure that all outputs are 100% compliant with HAL's governing manuals.

9.1 Multi-Stage Testing Approach

1. **Unit Testing:** Individual validation of the AI's parsing capabilities and the Rule Engine's logic gates.

2. **System Integration Testing (SIT):** Ensuring the smooth flow of data from the IFS ERP/CSV inputs through to the final consolidated RFQ ZIP package.
3. **Accuracy Verification:** A specific testing track where AI-extracted data is compared against "Ground Truth" data from 50+ historical HAL tenders.
4. **Security & Penetration Testing:** Onsite testing to verify that the application has no external leaks and follows HAL's internal data handling protocols.

9.2 User Acceptance Testing (UAT)

The final validation phase involves HAL's own procurement officers:

- **Scenario-Based Testing:** Users will process real-world requirements to verify if the tool correctly suggests MII (Make in India) clauses and commercial terms.
- **Excel Integrity Check:** Verifying that the output Excel files maintain all original HAL formulas, locked cells, and formatting.
- **Feedback Loop:** Any discrepancies found during UAT are addressed before the final production Go-Live

X. RISK MANAGEMENT

The system incorporates several "Fail-Safe" mechanisms to manage technical and operational risks:

- **The Deterministic Override:** If the AI engine suggests a clause that contradicts a mandatory policy rule (e.g., financial threshold rules), the **Rule Engine automatically rejects the suggestion** and notifies the user.
- **Environment Isolation:** To mitigate the risk of data theft or unauthorized access, the system remains strictly air-gapped from the public internet.
- **Version Control:** Every version of the Master Clause Library is archived. If a policy change is rolled back at HAL, the system can immediately revert to the previous governed version of the clauses.

XI. COMMERCIALS AND INVESTMENT

The investment model is designed to provide HAL with a total cost of ownership (TCO) advantage by utilizing open-source AI frameworks and a one-time onsite implementation fee, followed by long-term support.

- **Software Licensing:** No recurring per-user SaaS fees. The system utilizes locally hosted, open-source LLMs (Llama 3.2 / Mistral), ensuring no external licensing dependencies for the AI core.
- **Onsite Professional Services:** Covers the 28-week implementation cycle, including the deployment of the 5-person specialist team.
- **Hardware Procurement:** A one-time capital expenditure for the GPU-enabled server stack (as specified in Section VI) to be owned and maintained by HAL.
- **Long-Term Support (LTS):** A structured Annual Maintenance Contract (AMC) covering years 2 through 10, ensuring the tool remains compatible with evolving HAL IT infrastructure.

XII. MAINTENANCE AND SUPPORT (SLAs)

To ensure the long-term reliability and operational stability of the **Smart Clause Selection Tool** within the HAL MCSRDC environment, the project includes a comprehensive support framework. This framework is anchored by a minimum **10-year maintenance and technical support commitment**.

The following Service Level Agreements (SLAs) are designed to provide HAL with predictable, high-quality technical assistance throughout the system's lifecycle:

12.1 Incident Classification and Response Times

Incidents are categorized based on their impact on the procurement workflow, with corresponding response and resolution targets:

12.2 Support Delivery Model

- **Onsite Stabilization:** Following Phase 6 (Deployment), a dedicated **Deployment/MLOps Engineer** with 5+ years of experience will remain onsite to lead the stabilization phase and ensure the hardware is optimized for local AI inference.
- **Technical Documentation:** Support is backed by the handover of complete technical documentation and user manuals to HAL administrators and end-users during Phase
- **Preventative Maintenance:** Regular system health checks will be performed to ensure the local Vector Store (ChromaDB) and Relational Database (MySQL) remain performant and clear of data fragments.

12.3 Long-Term Sustainability

- **Policy Updates:** As HAL's procurement manuals and policy circulars evolve, the support team will assist authorized administrators in updating the **Master Clause Library** and the **deterministic Rule Engine** to maintain 100% compliance.
- **Model Maintenance:** The 10-year support plan includes periodic optimization of the local AI/NLP models to ensure they continue to accurately parse increasingly complex technical documents and natural-language queries.
- **Data Sovereignty Compliance:** All support activities, including bug fixes and library updates, will be conducted strictly within the **HAL intranet environment**, ensuring no data is ever exposed to external networks.

12.4 Training and Warranty

- A one-week onsite training program shall be provided for end-users and administrators, including hands-on sessions for complete system familiarization
- The system shall be installed and configured onsite at MCSRDC with full deployment support
- Technical support shall be provided to resolve issues encountered during system usage
- The complete software solution, including application modules, AI components, integrations, and databases, shall be covered under a comprehensive warranty for a period of 12 months following the post go-live support phase

XIII. COMPANY PROFILE

Our Vision

Leadership in architecture, design, development and deployment of mission critical applications in our key expertise domains: C4I systems for effective situational awareness and decision support, AI/ML based big data analytics, enterprise integration of real time data from IIOT and embedded/FPGA designs.

At C2C Advanced Systems, we are driven by an unwavering commitment to delivering value to our esteemed customers. We relentlessly pursue excellence, focusing on achieving unmatched proficiency in architecture, design, development, deployment, and through-life support within our key domains of expertise

A bird eye view of C2C Advanced System

C2C Advanced Systems commenced operations as DB Systems Inc. in the year 2011, as a company catering for the requirement of the Defence in general and Indian Navy in particular. The company's activities are performed by three main groups which are seamlessly integrated viz. Systems Engineering Group, Software Group and the Hardware Group.

The company joined the C2Ci group of companies in 2018, thus augmenting its financial and technical muscle enabling it to undertake large projects on a turn-key basis.

The Systems Engineering Group undertakes systems engineering activities like Systems Design, System Development, Systems Integration etc. This group is responsible for converting the user specifications and requirements into a consolidated system consisting of Hardware and Software. Experts who are part of this group, bring to bear their vast experience, in systems design and engineer a system that can be developed and integrated into a complete system.

This group is also responsible for integration of the software and hardware components

designed for the system and ensuring that the final product meets the customer requirements. We have been at the forefront of product development in mission critical applications. Our key expertise includes C4I systems for effective situational awareness and decision support, AI/ML based big data analytics, enterprise integration of real time data from IIOT and embedded/FPGA designs.

The Software group is responsible for design and development of software intensive systems, based on user requirements. The group consists of software engineers and Software testers who take up the task of design and development of software, following the industry standards. The Software Development Life Cycle followed is based on IEEE 12207 and all the artefacts are generated based on established software development templates.

The hardware group is responsible for design and development of hardware required as part of the complete system. The group consists of Mechanical and Electronics engineers well versed in mission critical hardware components in keeping with the latest industry standards. At C2C Advanced Systems, we understand that every client has unique requirements. Thus, we offer the flexibility to adapt our products and services to meet your specific needs. Whether you seek specialized skills to support technology creation or require comprehensive professional services that lead to managed solutions, we are here to help. We build with current generation technologies and continuously adopt future trends in our capabilities.

C2C-AS has Head Office in Bengaluru, Corporate Office at New Delhi and offices in Malaysia & US with 120+ employees in total. Through our presence we are covering the markets of India, Middle East, and Americas. We are Through onsite offshore model we are providing high quality services round the clock to global clientele.

Quality Assurance

Quality is the responsibility of each team member working at the company. C2C Advanced is a CMMI Level III registered company and our Quality Assurance (QA) Department is directly responsible for the revision and maintenance of all QA policies and procedures.

The QA Department is also responsible for ensuring each and every member of the project team follows all required policies and procedures. This ensures that quality is built in as part of the overall project, as opposed to merely checking for quality at the end of the production process.

A critical component of our CMMI compliance program includes a regular internal and external audit schedule. All departments at C2C Advanced are audited on a regular basis throughout the year, with external compliance audits scheduled yearly by an independent certification company. Corrective Action Reports (CARs) are prepared as required following internal and external reviews, to identify areas where process improvement is necessary.

TECHNICAL APPROACH AND VALIDATION REPORT

Industry 4.0 Smart Clause Selection

Prepared For (End Customer):

MCSRDC Division, Hindustan Aeronautics Limited
Marathalli post, Bengaluru 560037



Prepared By:

C2C Advanced Systems Limited
EPIP Zone, Whitefield,
Bengaluru 560066



Project / Reference Details:

Project Name: Industry 4.0 Smart Clause Selection Tool

Project ID / Reference No.:

Document Version: 3.0

Date of Submission:

5th Jan 2026

5. Two

Table of Contents

1.Introduction	Error! Bookmark not defined.
2.Problem Statement.....	3
3.Objectives of the Smart Clause	3
4.High-Level System Architecture	4
5.Detailed Architecture Description	4
5.1 Presentation Layer:	4
5.2 API & Authentication Layer:.....	4
5.3 Application & Rule Engine Layer:	5
5.4 Annexure Processing Layer	5
5.5 Data Layer	5
5.6 Output & Packaging Layer	5
6.User Interface Flow and Input Dependency	5
6.1 Page 1: Annexure-1 and Annexure-4 Inputs	5
6.2 Page 2: Annexure-2 and Annexure-3 Inputs:	6
6.3 Estimate-Based Dependency	7
6.4 Combined Output Generation	7
6.5 Smart Clause Data Flow Architecture	8
7.Validation of Inputs and Outputs:.....	9
7.1 Annexure - 1	9
7.2 Annexure - 2	10
7.3 Annexure - 3	13
7.4 Annexure - 4	14
8.Conclusion	15
9.POC Completion Status:	15

1. Introduction

This document presents the Technical Approach and Validation Report for the SmartClause Proof of Concept (PoC) developed for the HAL Procurement System. The PoC expands the earlier implementation, which was limited to Commercial Terms and Conditions for Domestic Bidders (Annexure-2A) and Foreign Bidders (Annexure-2B), to now include Technical Compliance Document (Annexure-1), Commercial Terms and Conditions (Annexure-2), Make In India - Purchase Preference Compliance (Annexure-3), and Price Bid Format (Annexure-4) within a single, unified workflow.

The objective of this document is to describe the system architecture, user interaction flow, annexure logic, and validation outcomes in a clear and implementation-oriented manner, aligned with the actual PoC execution.

2. Problem Statement

Preparation of RFQs in large procurement organizations involves identifying and assembling the correct set of contractual, commercial, and compliance clauses from multiple regulatory sources. Clause applicability depends on several parameters such as tender mode, tender type, estimated value, vendor category, and nature of goods or services.

In the current manual process, this activity is time-consuming and prone to inconsistencies and omissions. SmartClause addresses this challenge by introducing a deterministic, rule-driven system that automates clause selection and annexure generation, thereby improving consistency, traceability, and turnaround time.

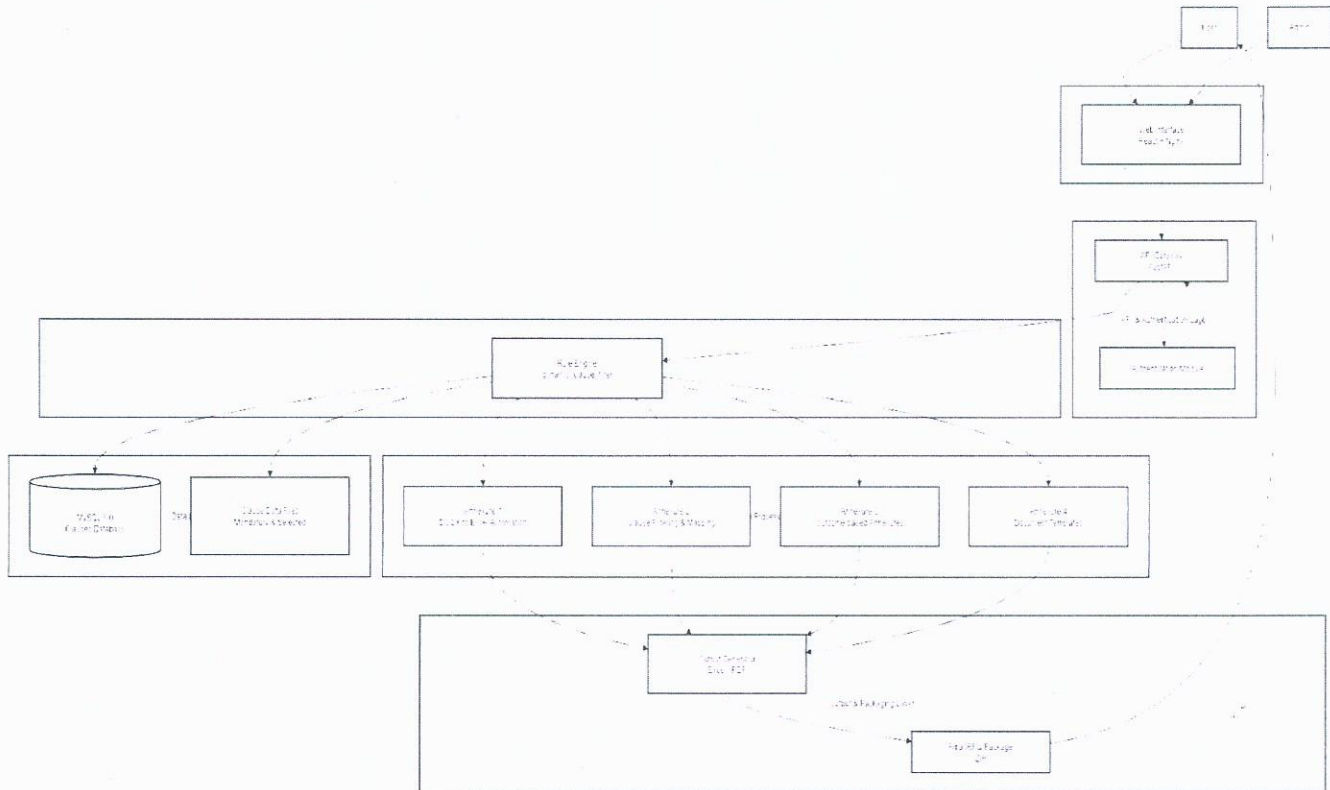
3. Objectives of the Smart Clause

The key objectives of the SmartClause are:

- To automate clause selection using deterministic, rule-based logic
 - To demonstrate end-to-end RFQ and annexure(s) generation
 - To validate clause applicability across different value thresholds and tender scenarios
 - To support offline deployment in a secure enterprise environment
 - To generate standardized Excel and PDF outputs that are directly usable by procurement users
-

4. High-Level System Architecture

The SmartClause PoC follows a layered and modular architecture to clearly separate user interaction, business logic, annexure processing, data management, and output generation. This design ensures maintainability, scalability, and ease of validation.



5. Detailed Architecture Description

5.1 Presentation Layer:

The presentation layer is a browser-based web application built using React and served through Nginx. It provides a guided, step-wise user experience for RFQ and annexure generation. The interface supports both standard users and administrative users, allowing administrators to manage clauses and configuration data.

5.2 API & Authentication Layer:

The backend is implemented using FastAPI and acts as the API gateway. This layer is responsible for request validation, authentication, and routing. The authentication mechanism follows the OAuth2 password flow standard, implemented using FastAPI's built-in security utilities. Upon successful credential verification, the server issues a JSON Web Token (JWT) that clients include in the Authorization header as a Bearer token for subsequent API requests.

JWTs are cryptographically signed using secure hashing algorithms (e.g., HS256) to ensure token integrity and prevent tampering. Only authenticated users with valid tokens are permitted to execute rule evaluations or generate annexures, ensuring controlled access within the secure enterprise environment.

5.3 Application & Rule Engine Layer:

This layer contains the core business logic of SmartClause. The Rule Engine, implemented in Python, evaluates tender input parameters against predefined deterministic rules. Based on this evaluation, the engine identifies applicable clauses and determines which annexures need to be generated.

5.4 Annexure Processing Layer

Each annexure is implemented as an independent module:

- **Annexure-1:** Automates conversion of proprietary DOCX documents into structured Excel compliance matrices.
 - **Annexure-2:** Filters and maps applicable clauses based on rule engine output.
 - **Annexure-3:** Generates outcome-based annexures driven primarily by estimated tender value.
 - **Annexure-4:** Generates pricing and document templates using uploaded files or external data sources.
-

5.5 Data Layer

The data layer consists of a MySQL 8.0 database storing master clauses, metadata, and rule mappings. Supporting data files are used for mandatory clause lists and annexure templates.

5.6 Output & Packaging Layer

This layer generates final Excel and PDF documents using standard libraries and packages all outputs into a single ZIP file for user download.

6. User Interface Flow and Input Dependency

The SmartClause PoC user interface is designed as a two-step workflow, clearly separating document-driven annexures from rule-driven annexures.

6.1 Page 1: Annexure-1 and Annexure-4 Inputs

The first page focuses on annexures that are independent of tender filtering parameters.

- **Annexure-1:** Users upload proprietary DOCX files, which are automatically converted into structured Excel compliance matrices. If an unsupported file format is uploaded (e.g., PDF, TXT), the system returns a clear error message indicating that only .docxfiles are accepted, prompting the user to upload a valid file.
- **Annexure-4:** Users upload pricing or data files (XLS/XLSX/CSV) or connect to an Oracle database to generate standardized pricing or document templates. The system validates incoming files against expected formats; uploads of unsupported types (e.g., images, PDFs) are rejected with an appropriate error message. Additionally, for database connections, the system handles connection failures gracefully by notifying the user of connectivity issues without exposing sensitive error details.

These inputs do not depend on tender mode, estimate, or vendor type.

The screenshot shows two side-by-side sections. The left section is titled 'Annexure 1 Generation' and contains a button that says 'Click to upload document'. The right section is titled 'Annexure 4 Generation' and contains a button that says 'Click to upload data file'. Both sections have a light blue header and a white body with a subtle grid pattern.

6.2 Page 2: Annexure-2 and Annexure-3 Inputs:

The second page captures tender-specific parameters required for clause selection:

- Mode of Tender
- Type of Tender
- Estimated Value (Estimate)
- Vendor Type
- Goods / Services

These parameters are passed to the Rule Engine for evaluation.

Step 2: Input Parameters

Mode of Tender

Type of Tender

Estimate

Others

Vendor Type

Goods/Services

6.3 Estimate-Based Dependency

The Estimated Value parameter plays a key role in output determination:

- It influences clause applicability in **Annexure-2**
- It directly controls outcome selection in **Annexure-3**

Specifically:

- Estimate < ₹10 Crores → Lower-value Annexure-3 output
- Estimate ≥ ₹10 Crores → Higher-value Annexure-3 output

Estimate

6.4 Combined Output Generation

After inputs from both pages are processed, outputs from Annexure-1 through Annexure-4 are generated in a single execution flow and packaged together as the final RFQ output.

Downloads > RFQ_Package_2025 12 17

Sort View Extract all

Annexure-1

Annexure-2

Annexure-3

Annexure-4

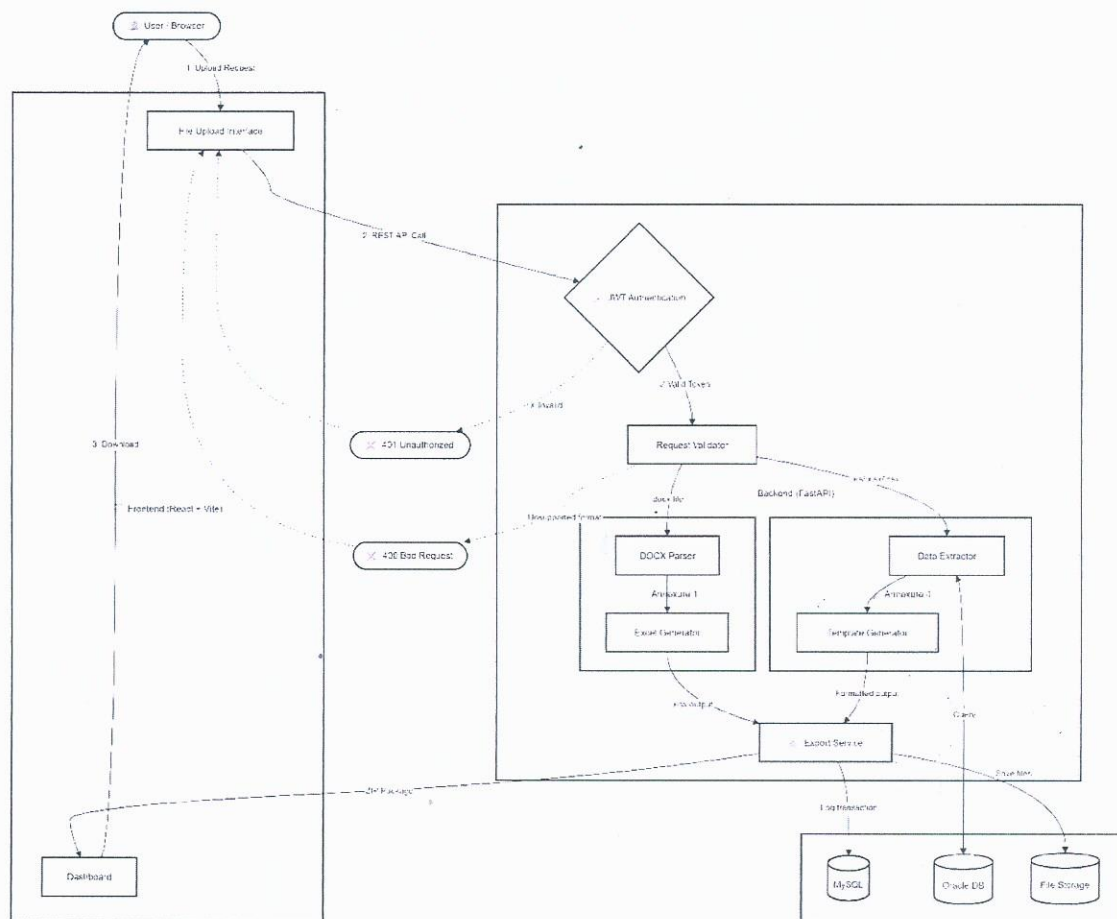
RFQ_Package_2025-12-17

18-12-2025 02:07

Compressed (zipp...

199 KB

6.5 Smart Clause Data Flow Architecture



7. Validation of Inputs and Outputs:

Validation was performed by correlating user inputs with system outputs across multiple test scenarios. The following observations were confirmed during the PoC:

- Identical inputs consistently produce identical outputs
- Annexure-3 outputs change correctly based on estimate thresholds
- Clause selection in Annexure-2 aligns with defined Input parameters
- Generated Excel and PDF files are structurally and functionally correct
- Zip file all Annexure Together

These results confirm the deterministic and reliable behavior of the SmartClause.

7.1 Annexure - 1

a) Input docx



TECHNICAL REQUIREMENT FOR PCB FABRICATION, COMPONENTS PROCUREMENT AND ASSEMBLY OF ELECTRONIC MODULES

Doc No: MCSRDC/HJT36/MC4/HW/FAB-ASSY/01/00
Date: 13th May, 2025

7.2 Annexure - 2

a) Test case – 1

Mode of Tender	Type of Vendor	Estimated Value	Others	Type of Vendor	Goods & Service

Expected output: Integrity Pact

Industry 4.0 Smart Clause Selection

Regist. Fee

Microbusiness Fee

Regist. Fee: The vendor must make payment to the Registrar for the registration fee which is the amount to be charged to the vendor for the right to make the registration of the tender. The fee is payable to the Registrar of the Government of Karnataka.

Microbusiness Fee: The vendor must make payment to the Registrar for the registration fee which is the amount to be charged to the vendor for the right to make the registration of the tender. The fee is payable to the Registrar of the Government of Karnataka.

Registration Fee: The vendor must make payment to the Registrar for the registration fee which is the amount to be charged to the vendor for the right to make the registration of the tender. The fee is payable to the Registrar of the Government of Karnataka.

Registration Fee: The vendor must make payment to the Registrar for the registration fee which is the amount to be charged to the vendor for the right to make the registration of the tender. The fee is payable to the Registrar of the Government of Karnataka.

Registration Fee: The vendor must make payment to the Registrar for the registration fee which is the amount to be charged to the vendor for the right to make the registration of the tender. The fee is payable to the Registrar of the Government of Karnataka.

b) Test case – 2

Mode of Tender	Type of Vendor	Estimated Value	Others	Type of Vendor	Goods & Service

Expected output: Security Deposit

Security Deposit

Design Terms: The terms of the tender shall be as per the Design Terms.

Services: The services to be provided shall be as per the Design Terms.

Design Terms: The terms of the tender shall be as per the Design Terms.

c) Test case – 3

Mode of Tender	Type of Vendor	Estimated Value	Others	Type of Vendor	Goods & Service

Expected output: PBG both Indigenous & Foreign bidders

S. T. S.

^a The number of subjects who were included in each group was 10.

$$\frac{\partial}{\partial t} \left(\frac{1}{2} \rho \mathbf{u} \cdot \mathbf{u} \right) + \nabla \cdot \left(\frac{1}{2} \rho \mathbf{u} \otimes \mathbf{u} \right) = \nabla \cdot \left(\frac{1}{2} \rho \mathbf{u} \otimes \mathbf{u} \right) + \nabla \cdot \left(\frac{1}{2} \rho \mathbf{u} \otimes \mathbf{u} \right)$$

Mode of Tender	Type of Vendor	Estimated Value	Others	Type of Vendor	Goods & Service

[illegible]

Mode of Tender	Type of Vendor	Estimated Value	Others	Type of Vendor	Goods & Service

Performance Bank Guarantee

[illegible]

Journal of Management Inquiry 18(6)

© 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026, 2027, 2028, 2029, 2030, 2031, 2032, 2033, 2034, 2035, 2036, 2037, 2038, 2039, 2040, 2041, 2042, 2043, 2044, 2045, 2046, 2047, 2048, 2049, 2050, 2051, 2052, 2053, 2054, 2055, 2056, 2057, 2058, 2059, 2060, 2061, 2062, 2063, 2064, 2065, 2066, 2067, 2068, 2069, 2070, 2071, 2072, 2073, 2074, 2075, 2076, 2077, 2078, 2079, 2080, 2081, 2082, 2083, 2084, 2085, 2086, 2087, 2088, 2089, 2090, 2091, 2092, 2093, 2094, 2095, 2096, 2097, 2098, 2099, 2100, 2101, 2102, 2103, 2104, 2105, 2106, 2107, 2108, 2109, 2110, 2111, 2112, 2113, 2114, 2115, 2116, 2117, 2118, 2119, 2120, 2121, 2122, 2123, 2124, 2125, 2126, 2127, 2128, 2129, 2130, 2131, 2132, 2133, 2134, 2135, 2136, 2137, 2138, 2139, 2140, 2141, 2142, 2143, 2144, 2145, 2146, 2147, 2148, 2149, 2150, 2151, 2152, 2153, 2154, 2155, 2156, 2157, 2158, 2159, 2160, 2161, 2162, 2163, 2164, 2165, 2166, 2167, 2168, 2169, 2170, 2171, 2172, 2173, 2174, 2175, 2176, 2177, 2178, 2179, 2180, 2181, 2182, 2183, 2184, 2185, 2186, 2187, 2188, 2189, 2190, 2191, 2192, 2193, 2194, 2195, 2196, 2197, 2198, 2199, 2200, 2201, 2202, 2203, 2204, 2205, 2206, 2207, 2208, 2209, 2210, 2211, 2212, 2213, 2214, 2215, 2216, 2217, 2218, 2219, 2220, 2221, 2222, 2223, 2224, 2225, 2226, 2227, 2228, 2229, 2230, 2231, 2232, 2233, 2234, 2235, 2236, 2237, 2238, 2239, 2240, 2241, 2242, 2243, 2244, 2245, 2246, 2247, 2248, 2249, 2250, 2251, 2252, 2253, 2254, 2255, 2256, 2257, 2258, 2259, 2260, 2261, 2262, 2263, 2264, 2265, 2266, 2267, 2268, 2269, 2270, 2271, 2272, 2273, 2274, 2275, 2276, 2277, 2278, 2279, 2280, 2281, 2282, 2283, 2284, 2285, 2286, 2287, 2288, 2289, 2290, 2291, 2292, 2293, 2294, 2295, 2296, 2297, 2298, 2299, 2300, 2301, 2302, 2303, 2304, 2305, 2306, 2307, 2308, 2309, 2310, 2311, 2312, 2313, 2314, 2315, 2316, 2317, 2318, 2319, 2320, 2321, 2322, 2323, 2324, 2325, 2326, 2327, 2328, 2329, 2330, 2331, 2332, 2333, 2334, 2335, 2336, 2337, 2338, 2339, 2340, 2341, 2342, 2343, 2344, 2345, 2346, 2347, 2348, 2349, 2350, 2351, 2352, 2353, 2354, 2355, 2356, 2357, 2358, 2359, 2360, 2361, 2362, 2363, 2364, 2365, 2366, 2367, 2368, 2369, 2370, 2371, 2372, 2373, 2374, 2375, 2376, 2377, 2378, 2379, 2380, 2381, 2382, 2383, 2384, 2385, 2386, 2387, 2388, 2389, 2390, 2391, 2392, 2393, 2394, 2395, 2396, 2397, 2398, 2399, 2400, 2401, 2402, 2403, 2404, 2405, 2406, 2407, 2408, 2409, 2410, 2411, 2412, 2413, 2414, 2415, 2416, 2417, 2418, 2419, 2420, 2421, 2422, 2423, 2424, 2425, 2426, 2427, 2428, 2429, 2430, 2431, 2432, 2433, 2434, 2435, 2436, 2437, 2438, 2439, 2440, 2441, 2442, 2443, 2444, 2445, 2446, 2447, 2448, 2449, 2450, 2451, 2452, 2453, 2454, 2455, 2456, 2457, 2458, 2459, 2460, 2461, 2462, 2463, 2464, 2465, 2466, 2467, 2468, 2469, 2470, 2471, 2472, 2473, 2474, 2475, 2476, 2477, 2478, 2479, 2480, 2481, 2482, 2483, 2484, 2485, 2486, 2487, 2488, 2489, 2490, 2491, 2492, 2493, 2494, 2495, 2496, 2497, 2498, 2499, 2500, 2501, 2502, 2503, 2504, 2505, 2506, 2507, 2508, 2509, 2510, 2511, 2512, 2513, 2514, 2515, 2516, 2517, 2518, 2519, 2520, 2521, 2522, 2523, 2524, 2525, 2526, 2527, 2528, 2529, 2530, 2531, 2532, 2533, 2534, 2535, 2536, 2537, 2538, 2539, 2540, 2541, 2542, 2543, 2544, 2545, 2546, 2547, 2548, 2549, 2550, 2551, 2552, 2553, 2554, 2555, 2556, 2557, 2558, 2559, 2560, 2561, 2562, 2563, 2564, 2565, 2566, 2567, 2568, 2569, 2570, 2571, 2572, 2573, 2574, 2575, 2576, 2577, 2578, 2579, 2580, 2581, 2582, 2583, 2584, 2585, 2586, 2587, 2588, 2589, 2590, 2591, 2592, 2593, 2594, 2595, 2596, 2597, 2598, 2599, 2600, 2601, 2602, 2603, 2604, 2605, 2606, 2607, 2608, 2609, 2610, 2611, 2612, 2613, 2614, 2615, 2616, 2617, 2618, 2619, 2620, 2621, 2622, 2623, 2624, 2625, 2626, 2627, 2628, 2629, 2630, 2631, 2632, 2633, 2634, 2635, 2636, 2637, 2638, 2639, 2640, 2641, 2642, 2643, 2644, 2645, 2646, 2647, 2648, 2649, 2650, 2651, 2652, 2653, 2654, 2655, 2656, 2657, 2658, 2659, 2660, 2661, 2662, 2663, 2664, 2665, 2666, 2667, 2668, 2669, 2670, 2671, 2672, 2673,

f) Test case – 6

Mode of Tender	Type of Vendor	Estimated Value	Others	Type of Vendor	Goods & Service

Expected output: PBG Foreign bidder

Performance Bank Guarantee

Design and Development of a new product for the purpose of the project.

Sub-clause 1

Performance Bank Guarantee - Foreign

g) Test case – 7

Mode of Tender	Type of Vendor	Estimated Value	Others	Type of Vendor	Goods & Service

Expected output: Withholding Tax

Withholding Tax

7.3 Annexure - 3

a) Input from annexure 2 parameters

Estimate

5 Crore

5 Lakhs

10 Lakhs

50 Lakhs

1 Crore

10 Crore

S. T. S.

b) Output Generated

MISSION & COMBAT SYSTEM RESEARCH & DESIGN CENTRE
HINDUSTAN AERONAUTICS LIMITED
Marathalli Post
Bangalore - 560 037 INDIA

Public Procurement (Preference to Make in India) - Part of Technical Bid - Annexure-3

NAME OF WORK: Workstation for Simulators

NAME OF THE VENDOR/FIRM:

QUOTATION REF. NO.:

PLEASE INDICATE YOUR RESPONSE/ COMPLAINT/ ACCEPTANCE TO THE FOLLOWING TERMS (MANDATORY) AND UPLOAD AS A PART OF TECHNICAL BID.

In line with Public Procurement (Preference to Make in India), Order 2017 & its amendments issued by Govt. of India & with a view to support the Indian industries, HAL has implemented "Purchase Preference Policy". The "Purchase Preference" is applicable for the "Class-I Local Supplier" for the goods/services/works covered in this tender subject to the following terms & conditions:

Note:

(a) The supplier must fulfill under the following conditions:

(b) The orders bought only from: **Class-I & Class-II local suppliers** out of local suppliers.

Exemptions: A supplier or service provider, whose goods, services or works referred for procurement, has local content:

a. Equal to or more than 50% : **Class-I local supplier**

b. More than 20% but less than 50% : **Class-II local supplier**

c. Less than or equal to 20% : **Non local supplier**

Sl.	HAL, RFQ terms & conditions	Vendor Compliance Remarks
1	a. Equal to or more than 50% : Class-I local supplier .	
2	b. More than 20% but less than 50% : Class-II local supplier .	
3	Local content means the amount of value added in India (i.e. indigenous items, services added in the other products, services, works, be the total value of the item referred, including not considering indirect taxes minus the value of imported content in the item including all indirect taxes (GST) as a proportion of the total value (including not domestic indirect taxes) in the item.	

7.4 Annexure - 4

a) Input csv

Line No.	Description	Quantity	Purch. U/N	Wanted R/L	Latest Und	Supplier	Supplier's Supplier	U-Status	Buyer	Request to Formulate	Note Text	Document	Price Unit	Currency	Estimated Price/Desc	Discount	Additional	Total Price	Total	Sub
1	Fabricated	8	1	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
2	NRE for fa	1.00	*****	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
3	Group-B R	1.00	*****	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
4	Kit building	8.00	*****	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
5	Component	2	2	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
6	IOCM Assy	8	1	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
7	NRE Chang	1.00	*****	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
8	Conforma	8.00	*****	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
9	IOCM Assy	5.00	*****	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
10	Fabricated	8	1	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
11	NRE for fa	1.00	*****	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
12	Group-B R	1.00	*****	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
13	IOCM Assy	8	1	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
14	NRE Chang	1.00	*****	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
15	Conforma	8.00	*****	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
16	IOCM Assy	5.00	*****	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
17	Fabricated	8	1	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
18	NRE for fa	1.00	*****	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
19	Group-B R	1.00	*****	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
20	Kit building	8.00	*****	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
21	Component	2	2	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
22	IOCM Assy	8	1	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
23	NRE Chang	1.00	*****	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
24	Conforma	8.00	*****	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
25	IOCM Assy	5.00	*****	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
26	Fabricated	8	1	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
27	NRE for fa	1.00	*****	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84
28	Group-B R	1.00	*****	*****	*****	17008347	Deccel Electronics De PO Create *			Request to Formulate	FALSE	0	0 INR	CORRECT	0	0	0	0	0	0.84

b) Output Generated

ANNEXURE-4

Mission & Combat System Research & Design Centre
HINDUSTAN AERONAUTICS LIMITED
Marathahalli Post
Bangalore - 560 037 INDIA

Commercial Bid (PRICE FORMAT)

Tender Ref No: I/C S461 2238 25-1

Type of Bid: Two Bid System

PCB FABRICATION, COMPONENTS PROCUREMENT AND ASSEMBLY OF ELECTRONIC MODULES

Name of the Vendor Firm:

Quotation Ref No:

SLNo.	Part Description	Qty	(UOM)	HSN/SAC Code	Currency	Basic Unit Cost	Sub Total (Basic Unit Cost X Qty.)	GST (%)	GST Amount	Total line value (Sub Total + GST Amount)
A. Phase-1 (Design and Development Phase)										
1	Fabricated MIL Grade IOCM PCB along with Group-A report and CoC.	2	1							
2	LIRE for fabrication of MIL Grade IOCM PCB	1	JB							
3	Group-B Report for Fabricated MIL Grade IOCM PCB	1	JB							

8. Conclusion

The SmartClause PoC successfully demonstrates a complete, rule-driven approach for RFQ clause selection and annexure generation. By integrating Annexure-1, Annexure-2, Annexure-3, and Annexure-4 into a unified workflow, the system significantly reduces manual effort, improves compliance consistency, and provides a scalable foundation for future enhancements.

Deployment & Testing Summary:

The PoC has been successfully deployed on a production server using Dockerized services (MySQL, FastAPI backend, Nginx frontend) and tested across multiple client systems connected to the server. All test scenarios produced correct outputs, validating the system's reliability in a networked enterprise environment.

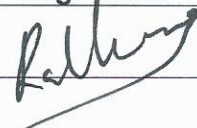
9. POC Completion Status:

Completed and report submitted to End customer. Details below:

For Technical Team

Sl. No.	Name	Designation	Signature	Date
1	Annapurneshwari Battur	Jr. AI Engineer	A.M. Battur	06/01/26
2	Jagadeesh Reddy R	Data Scientist	Jagadeesh Reddy R	07/01/26

For Service Provider / Supplier

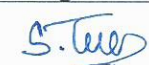
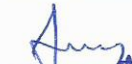
Sl. No.	Name	Designation	Signature	Date
1	RAVIKANTH	SVP		06/Jan/26

Document Status:

☒ Approved ☐ Approved with Comments ☐ Rejected

Remarks (if any):

For End Customer

Sl. No.	Name	Designation	Signature	Date
1	THEJAS. S	SM(Purchase)		06 th Jan 2026
2	A. KANNAN	DCM(Imm)		06-01-26
3	Siva Pavan Kumar	CM(D)		06/Jan/26

----- END OF FILE -----